ACID Properties in MySQL

1. **Atomicity**:
    - In MySQL, transactions are atomic, meaning all operations within a transaction are treated as a single unit. If any part of the transaction fails, the entire transaction fails, and no changes are made to the database. You can use the `BEGIN`, `COMMIT`, and `ROLLBACK` statements to manage transactions.

Example:

```
START TRANSACTION;
UPDATE accounts SET balance = balance - 100 WHERE account_id = 1;
UPDATE accounts SET balance = balance + 100 WHERE account_id = 2;

-- If there's an error, roll back changes
ROLLBACK;  -- Undo all changes if something goes wrong

-- If everything is fine
COMMIT;    -- Save all changes
```

2. **Consistency**:
    - MySQL ensures that a transaction brings the database from one valid state to another. All predefined rules (like foreign key constraints and data types) must be followed.

Example:

```
-- Suppose there’s a rule that an account cannot have a negative
balance.

UPDATE accounts SET balance = balance - 100 WHERE account_id = 1;

-- If this causes a negative balance, the database will reject the
transaction.
```

3. **Isolation**:

- MySQLIsolation ensures that when multiple transactions happen at the same time, they don't interfere with each other. Each transaction works as if it's the only one running.

4. **Durability**:
    - Once a transaction has been committed in MySQL, the changes are permanent, even in the case of a crash. MySQL uses a transaction log to ensure that committed transactions are not lost.

Example:

```
-- After a COMMIT, the changes are saved
COMMIT;

-- Even if the server crashes after this point, the changes will
remain.
```